SNIGDHA CHANDRATRE

Data Science • Machine Learning • Deep Learning

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PROFESSIONAL SUMMARY

Data enthusiast with experience in machine learning and deep learning. Interested in using data-driven insights to solve complex real-world challenges across various domains. Published researcher with a foundation in developing deep learning algorithms. Seeking an opportunity to work in a dynamic, data-driven environment.

TECHNICAL SKILLS

- ✓ **Languages & Libraries:** Python (NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow) • SQL
- ✓ **ML / DL:** Supervised & Unsupervised Learning • CNNs • RNNs • LSTM • GRU • Transformers • SVM • Decision Trees • Random Forests
- ✓ **Specializations:** Clustering & Classification Algorithms • Feature Engineering • Data Visualization (Matplotlib, Seaborn)
- ✓ **Tools:** Jupyter Notebooks • MS Excel • MATLAB/Simulink

PUBLICATIONS

- ✓ Pankaj Warule, **Snigdha Chandratre,** S. Daware, et al., “Dual-Tree Complex Wavelet Transform for the Automatic Detection of the Common Cold Based on Speech Signals,” *Circuits, Systems, and Signal Processing* (2025). DOI
- ✓ Pankaj Warule, **Snigdha Chandratre,** Siba Prasad Mishra, Suman Deb, “Detection of the Common Cold from Speech Signals Using Transformer Model and Spectral Features,” *Biomedical Signal Processing and Control*, Volume 93, 2024, 106158. DOI
- ✓ **Snigdha Chandratre,** Pankaj Warule, Siba Prasad Mishra, Suman Deb, (2024). “LSTM- and GRU-Based Common Cold Detection from Speech Signal.” In: Hirose, K., Joshi, D., Sanyal, S. (eds) *Proceedings of 27th International Symposium on Frontiers of Research in Speech and Music. FRSM 2023.* Advances in Intelligent Systems and Computing, vol 1455. Springer, Singapore. DOI

KEY PROJECTS

CNN-Based Handwritten Digit Recognition — Edge Deployment

- ✓ **Achieved 98.76% accuracy** on MNIST hand-written digits dataset using a custom CNN (TensorFlow), and used quantization to **shrink model size by 70%**.
- ✓ Deployed on Android for fully **on-device inference** with no cloud dependency.
- ✓ Github Project link

Common Cold Detection from Speech Signals — Published Research

- ✓ **Outperformed all prior baselines** with a Transformer-based classifier achieving 69% accuracy on voice signal disease detection.
- ✓ Benchmarked SVM, Decision Trees, Random Forests, DNNs, LSTM, and GRU — findings published in **Biomedical Signal Processing and Control (2024)**.
- ✓ Evaluated models using Accuracy, Precision, Recall and F1-score across all classifiers to select the best-performing architecture.

Diwali Sales Analysis | Python, Pandas, Seaborn, Matplotlib

- ✓ Performed **EDA** and data preprocessing on 11K+ retail transactions to identify customer segments and revenue drivers.
- ✓ Identified **key revenue drivers across demographics, geography, and product categories**
- ✓ Translated data into actionable business insights to support marketing using visualizations.
- ✓ Github Project link

Torque Model Prediction for Vehicle ECU — Deep Learning

- ✓ **Exceeded static model baseline** (R^2 = 0.7) by designing dynamic deep learning models achieving **R^2 = 0.8** for torque prediction in ECU simulations.

Agricultural Leaf Disease Detection — Computer Vision

- ✓ Built an end-to-end leaf disease classification system using K-Means clustering and image processing for automated agricultural infection monitoring.

WORK EXPERIENCE

Senior Engineer | Bosch (BGSW), Bengaluru *Sep 2023 – Oct 2024*

- ✓ **Performed ECU Calibration for vehicle and** developed data driven models for vehicle performance prediction using deep learning techniques.
- ✓ Applied feature extraction and data analysis in MATLAB and Simulink to enhance system diagnostics.
- ✓ **Recognized with an Appreciation Certificate** for leading a complex technical software migration project.

M.Tech. Intern | Bosch (BGSW), Bengaluru *Jun 2022 – Apr 2023*

- ✓ **Designed and validated LSTM/GRU models** for vehicle signal time-series analysis, benchmarking architectures for ECU applications.
- ✓ **Built a MATLAB R^2 Calculator GUI tool** adopted by the team to standardize model evaluation and accelerate signal comparison.

Associate Software Engineer | Accenture, Pune

- ✓ Revised, modularized and updated old code bases to modern development standards, reducing operating costs and improving functionality.
- ✓ Worked in SAP ABAP and SAP PI in the simultaneously to co-ordinate between the two modules.

EDUCATION

M.Tech., Communication Systems — NIT (SVNIT), Surat *2021 – 2023*

- ✓ GPA: 8.22/10 • **Perfect 10/10 CGPA in Semester 4** • Thesis: Deep learning for speech-based disease detection.

B.E., Electronics & Telecommunications — NBN Sinhgad School of Engineering, Pune *2014 – 2018*

ACHIEVEMENTS

- ✓ **Appreciation Certificate** from Technical Project Manager at Bosch — recognized for leading a complex software migration project.
- ✓ **Conference Presenter** — FRSM 2023 (Springer), presented research on LSTM/GRU-based disease detection from speech.
- ✓ **Academic Excellence** — Perfect 10/10 CGPA in M.Tech. Semester 4, SVNIT Surat.